

Network Security Project

DNS Cache Poisoning and DNSSEC Defense

Overview

In this project, students will simulate a DNS cache poisoning attack and then implement DNSSEC to defend against it. The project highlights how DNS vulnerabilities can be exploited and how cryptographic protections mitigate such attacks.

Learning Objectives

- Understand DNS resolution and recursive resolvers
- Implement a DNS cache poisoning attack
- Understand bailiwick checking and DNS weaknesses
- Deploy and validate DNSSEC
- Analyze attack vs defense effectiveness

Environment Setup

- Attacker machine (Kali Linux recommended)
- DNS resolver (Bind or Unbound)
- Victim client machine
- Wireshark for traffic analysis

Project Tasks

Part 1: DNS Cache Poisoning

1. Set up a local DNS resolver (Bind/Unbound)
2. Configure the resolver to allow queries from the victim
3. Launch spoofed DNS responses targeting the resolver
4. Inject a fake A record into the cache
5. Demonstrate redirection of victim traffic to malicious IP

Part 2: Attack Analysis

6. Capture traffic using Wireshark
7. Analyze transaction IDs and ports
8. Explain why the attack succeeds

Part 3: DNSSEC Deployment

9. Enable DNSSEC validation on resolver
10. Configure trust anchors
11. Repeat the attack attempt
12. Show that poisoning fails

Expected Deliverables

- Working attack demonstration
- DNSSEC validation screenshots
- Traffic analysis
- Demo video

Rules:

- The project is to be done in teams of 2 – 3 students.
- Submissions: Github repo of the code, technical report detailing the setup and screenshots of the results, demo video.
- Deadline to submit the deliverables: 12 May, 11:59pm.
- Demo presentations will be conducted in the last class of the semester, 14 May.
- AI use is allowed.